Solutions to the Assignment for Week 2

Median voter convergence, logrolling, signalling, and the politicians' wage

Jacob Edenhofer*

14/05/2026

Remark. *These are model answers, not the only correct ones. The simulation code is in R; equivalent Python (`numpy/matplotlib`) is straightforward. For the essay questions I sketch the argument structure I would expect; depth and examples are entirely up to you. Equilibrium analyses in §3 and §4 use the standard Perfect Bayesian Equilibrium (PBE) refinement throughout.*

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1 The Downsian model and median voter convergence

1.1 Convergence and uniqueness (formal)

(a) Why $x_A < x_B$ with $x_A < m$ is not an equilibrium. A pure-strategy Nash equilibrium is a profile (x_A^*, x_B^*) such that neither party can strictly increase its vote share by unilaterally

*jacob.edenhofer@nuffield.ox.ac.uk

deviating. Suppose $x_A < x_B$ and $x_A < m$. Voters with ideal points below the midpoint $\hat{x} = (x_A + x_B)/2$ vote for A; the rest vote for B. So A's vote share is $F(\hat{x})$. Consider the deviation $x'_A = x_A + \varepsilon$ for $\varepsilon > 0$ small. The new midpoint is $\hat{x}' = \hat{x} + \varepsilon/2 > \hat{x}$, and

$$F(\hat{x}') - F(\hat{x}) > 0$$

by strict monotonicity of F on the interval $[\hat{x}, \hat{x}']$. Hence A strictly gains, and the original profile is not an equilibrium. The strict-monotonicity assumption is exactly what rules out a flat segment of F around the midpoint, on which a small move would have no effect.

(b) (m, m) is the unique pure-strategy NE. At (m, m) both parties are tied at $1/2$ each (under symmetric tie-breaking). A deviation by A to $x'_A < m$ moves the midpoint to $(x'_A + m)/2 < m$; A's vote share becomes $F((x'_A + m)/2) < F(m) = 1/2$, so A strictly loses. Symmetrically for $x'_A > m$. Hence (m, m) is a NE.

For uniqueness: by part (a), no profile with $x_A \neq x_B$ can be an equilibrium. Consider $x_A = x_B = x \neq m$. WLOG $x < m$. Either party can deviate to $x' \in (x, m]$ and capture all voters with ideal points above the new midpoint $(x + x')/2$, giving vote share $1 - F((x + x')/2) > 1/2$ (since $(x + x')/2 < m$). So this is not an equilibrium either. The unique NE is (m, m).

(c) Voters who are indifferent abstain. At any profile with $x_A = x_B$, every voter is indifferent and abstains. With zero turnout the "vote share" is undefined; any tie-breaking convention will do, but the strategic structure is broken. In particular (m, m) ceases to be a strict NE: any ε-deviation by either party gives them positive vote share against zero for the rival. The deviator's best response is then to come back closer; no PSNE exists in this game. Mixed-strategy equilibria do, but at the cost of pinning down the median voter as a unique focal point.

(d) Policy-motivated A, vote-maximising B. With B at m, A's expected utility from choosing x_A is

$$\mathbb{E}[u_A(x_A)] = \pi_A(x_A)\, u_A(x_A) + (1 - \pi_A(x_A))\, u_A(m),$$

where $\pi_A(x_A) = F((x_A + m)/2)$ and $u_A(x) = -|z_A - x|$. For any $x_A \neq m$, $\pi_A(x_A) < 1/2$, so A loses with probability strictly above $1/2$ and the implemented policy is m in expectation. With $z_A < m$, locating at $x_A < m$ *would* bring $u_A(x_A) > u_A(m)$ when A wins, but A also wins less often.

If we add a small office tie-breaker (a strict preference for being elected when otherwise indifferent), $x_A = m$ is the unique best response: it secures the $1/2$ tied probability of office, while any $x_A \neq m$ both reduces π_A and produces no policy gain (since B wins and implements m). *Without* that tie-breaker, A is weakly indifferent across all x_A since the ultimate policy is always m.

If both parties are policy-motivated with $z_A < m < z_B$, full convergence *need not* obtain. Roemer's (2001) Party-Unanimity Nash equilibrium and the Wittman–Calvert tradition show that with policy preferences and no uncertainty, parties may diverge toward their ideal points; uncertainty about the median's location is what drags them back toward m.

1.2 Visualising convergence and its breakdown (simulation)

Setup. The best-response dynamic alternates: given x_B, set $x_A = \arg\max_x F((x + x_B)/2)$; given x_A, set $x_B = \arg\max_x 1 - F((x_A + x)/2)$. With strictly increasing F, both best responses are interior solutions converging on m.

Listing 1: Best-response iteration

```
br_dynamic <- function(F_inv, x0 = c(0.1, 0.9), tol = 1e-4, max_iter = 200) {
```

```
  x <- matrix(x0, nrow = 1)
  for (t in 1:max_iter) {
    xA <- min(F_inv(0.5), x[t,2])  # A wants midpoint at median
    xB <- max(F_inv(0.5), xA)
    x <- rbind(x, c(xA, xB))
    if (max(abs(x[t+1,] - x[t,])) < tol) break
  }
  x
}
```

(a) $F = \text{Uniform}[0, 1]$**.** Empirical median $m \approx 0.5$. From $(0.1, 0.9)$ both parties' best response is the median; convergence in one iteration. Trajectory: $(0.1, 0.9) \to (0.5, 0.5)$.

(b) $F = \text{Beta}(2, 5)$**.** Density is right-skewed; analytic median is ≈ 0.265. The empirical median from $N = 10{,}000$ draws is in the same neighbourhood. Equilibrium positions converge to $(\hat{m}, \hat{m})$ with $\hat{m} \approx 0.265$. Note: *convergence speed* depends only on the dynamic, not on F; under exact best response it is one step. Iterative best-response with smoothing or noise would slow convergence.

(c) Bimodal mixture. The mixture $0.5\mathcal{N}(0.3, 0.05^2) + 0.5\mathcal{N}(0.7, 0.05^2)$ has median exactly 0.5 by symmetry. Both starting points still converge to $(0.5, 0.5)$. The substantive lesson: bimodal distributions *do not break* the MVT — the median voter still pins down convergence — but they do imply that both parties locate at a position that is far from where the population mass actually lies. This is the "hollow centre" phenomenon and motivates models with abstention or activist mobilisation.

(d) Abstention when indifferent. At (m, m) all voters abstain and both parties get zero votes. Any small deviation by A is strictly profitable. The dynamic enters a pattern with no fixed point: A deviates to $m - \varepsilon$, B's best response is to come back to slightly less than $m - \varepsilon/2$, etc. In the simulation you should observe oscillation rather than convergence to a single point.

(e) Tabulation.

F	m	x_A^*	x_B^*
Uniform$[0, 1]$	0.50	0.50	0.50
Beta$(2, 5)$	0.265	0.265	0.265
Bimodal mixture	0.50	0.50	0.50

The parties' joint distance from the population mean is zero in the symmetric cases and equals $|\mathbb{E}[X] - m|$ in the skewed case. The MVT cares about the median, not the mean.

1.3 Scope conditions and the puzzle of persistent divergence (essay sketch)

Argument structure I would expect. A strong essay states the assumption precisely, identifies the proof step it underwrites, and offers a sharply-bounded empirical assessment. Examples of good moves:

- *Two-party assumption* (contextual). Strategic role: closes off the "squeeze" geometry. Australia: nominally two-bloc, but the Coalition is itself a fusion of two parties (Liberal + National) with internal rents from divergence, and the Greens + minor parties absorb $\sim 12\%$ of the primary vote. The *effective* number of parties is closer to 3.

- *Single-peaked preferences* (demand-side). Strategic role: ensures the median voter's bliss point is the sole ideal-point that maximises the median voter's utility. Failure modes: salience-driven preferences over multiple issues can produce non-single-peaked induced preferences.
- *Voters care only about policy proximity* (demand-side). Strategic role: removes valence as a dimension. Failure mode: governing competence, leader effects, or descriptive representation. Note that this assumption *is* relaxed in §1 task 4.
- *Vote-maximising parties* (supply-side). Strategic role: the parties' best response is policy-positioning. Failure mode: policy-motivated factions or activists, who may pull the party away from m.
- *Costless platform mobility* (supply-side). Strategic role: parties can locate anywhere. Failure mode: reputational stickiness (Tavits 2007).

Assessing Australia. A serious answer notes that the most binding scope-condition violations seem to be (i) policy-motivated factions inside both major parties, (ii) compulsory voting, which removes the abstention margin and so makes activist mobilisation matter less than in optional-voting systems, and (iii) preferential ballots which sustain the viability of niche parties. Reasonable students conclude that Australian divergence is a *confirmation* of the model's scope conditions (specifically: that the single-dimension and unitary-vote-maximising-actor assumptions are violated), not a refutation of the underlying logic.

1.4 Valence and one-sided convergence

(a) Indifferent voter with valence. Take $x_A < x_B$. For voters with $x_i \in [x_A, x_B]$:

$$U_i(A) = -(x_i - x_A) + v_A, \quad U_i(B) = -(x_B - x_i) + v_B.$$

Indifference $U_i(A) = U_i(B)$ gives

$$-(x_i - x_A) + v_A = -(x_B - x_i) + v_B \iff \hat{x} = \frac{x_A + x_B}{2} + \frac{\Delta v}{2}, \quad \Delta v \equiv v_A - v_B.$$

With $\Delta v > 0$ the indifferent voter shifts *toward* B, so more voters strictly prefer A. Outside the interval $[x_A, x_B]$ the absolute-value forms flip; you can verify the same expression holds whenever $|\Delta v| \leq |x_B - x_A|$, which is the relevant range.

(b) Both parties co-located. If $x_A = x_B$, then $U_i(A) - U_i(B) = v_A - v_B = \Delta v > 0$ for every voter, so A wins all voters. The high-valence party can take any policy position the low-valence party takes and still win. This is what is meant by "flexibility": A need not match B position-for-position to win; A can credibly threaten to track B everywhere.

(c) Policy-motivated A, vote-maximising B at $m = 0.5$. With voters $\sim$ Uniform$[0, 1]$ and $\hat{x} = (x_A + 0.5 + \Delta v)/2$, A's vote share is $\hat{x}$. A wins (i.e. obtains a majority) iff $\hat{x} \geq 1/2$, i.e.

$$x_A \;\geq\; \tfrac{1}{2} - \Delta v.$$

Subject to this constraint, A minimises $|x_A - 0.3|$. For $\Delta v \leq 0.2$ the constraint binds, and the optimum is $x_A^* = \frac{1}{2} - \Delta v$, giving

$$|x_A^* - x_B^*| \;=\; \Delta v.$$

For $\Delta v > 0.2$ the constraint slackens and A locates at its bliss $x_A^* = 0.3$, with maximal divergence capped at 0.2.

(d) Third Way and the empirical content of convergence. Sketch. In the Third-Way era (Blair, Schröder), centre-left parties accepted much of the centre-right's economic programme. Two readings: (i) a Downsian story (centre-left parties chased the median voter on economics); (ii) a valence-substitution story (centre-left parties differentiated on perceived governing competence and managerial professionalism, conceding policy similarity in exchange for valence advantage). The model in (c) supports the second reading: when $\Delta v > 0$, observed convergence on x overstates the policy convergence implied by Downsian logic, because the high-valence party is *free* to converge without losing.

2 Logrolling and parties as institutional commitment devices

2.1 Spatial logrolling: who gains, who loses?

(a) Utilities at the median outcome. With $L = (0.2, 1.0)$, $M = (0.5, 0.5)$, $R = (1.0, 0.3)$ and $\alpha = 1/2$ for all,

$$U_L(0.5, 0.5) = -[(0.3)^2+(0.5)^2] = -0.34, \quad U_M(0.5, 0.5) = 0, \quad U_R(0.5, 0.5) = -[(0.5)^2+(0.2)^2] = -0.29.$$

(b) Utilities at $(0.6, 0.65)$**.**

$$U_L = -[0.16 + 0.1225] = -0.2825, \quad U_M = -[0.01 + 0.0225] = -0.0325, \quad U_R = -0.2825.$$

L improves from -0.34 to -0.2825; R improves from -0.29 to -0.2825; M falls from 0 to -0.0325. So L and R both gain, but M is harmed.

(c) "Pareto-superior" as misuse. A Pareto improvement requires *no* agent to be worse off. Since M is harmed, the bundle is not Pareto-superior to the median outcome. A precise label is *coalitional Pareto improvement (for the* $\{L, R\}$ *blocking coalition)*, or simply: *L and R are jointly better off, at M's expense.* The lecture's loose phrasing nonetheless captures something useful: in coalitional bargaining problems, the relevant solution concept is gains realisable by a winning coalition, not full Pareto improvement.

(d) Sum-maximising bundles. Maximising $U_L + U_R$ over $(x, y) \in \mathbb{R}^2$:

$$\frac{\partial}{\partial x}\left[-(x - 0.2)^2 - (x - 1.0)^2\right] = 0 \;\Rightarrow\; x^{**} = \tfrac{0.2+1.0}{2} = 0.6.$$

Symmetrically $y^{**} = (1.0 + 0.3)/2 = 0.65$. So $(0.6, 0.65)$ is exactly the slide bundle: it is the L–R utilitarian optimum.

For $U_L + U_M + U_R$:

$$x = \tfrac{0.2+0.5+1.0}{3} \approx 0.567, \quad y = \tfrac{1.0+0.5+0.3}{3} = 0.6.$$

The three-way utilitarian outcome differs because M's ideal pulls x toward 0.5 and y toward 0.5; the L–R coalition optimum simply ignores M.

2.2 Salience-based logrolling

(a) Default utilities. $U_L^{\mathrm{MR}} = -[0.3 \cdot 0.09 + 0.7 \cdot 0.25] = -[0.027 + 0.175] = -0.202$; $U_R^{\mathrm{MR}} = -[0.7 \cdot 0.25 + 0.3 \cdot 0.09] = -0.202$.

(b) Diagonal symmetry. With $\alpha_L + \alpha_R = 1$ and ideal points $L = (0.2, 1.0)$, $R = (1.0, 0.2)$, evaluate each U_i on the diagonal $x = y$:

$$U_L(x,x) = -[\alpha_L(x-0.2)^2 + (1-\alpha_L)(x-1)^2] = -[0.3(x-0.2)^2 + 0.7(x-1)^2],$$

$$U_R(x,x) = -[\alpha_R(x-1)^2 + (1-\alpha_R)(x-0.2)^2] = -[0.7(x-1)^2 + 0.3(x-0.2)^2] = U_L(x,x).$$

So along the diagonal $U_L = U_R$ exactly, and the bargaining problem collapses to a single-variable optimisation.

(c) Diagonal optimum. Differentiating $U_L(x,x)$:

$$\frac{dU_L(x,x)}{dx} = -[0.6(x-0.2) + 1.4(x-1)] = -(2x - 1.52).$$

Setting to zero gives $x^* = 0.76$. Plugging in:

$$U_L(0.76, 0.76) = -[0.3 \cdot (0.56)^2 + 0.7 \cdot (0.24)^2] = -[0.3 \cdot 0.3136 + 0.7 \cdot 0.0576] = -0.1344. \checkmark$$

(d) No salience asymmetry. With $\alpha_L = \alpha_R = 0.5$ and the same ideal points, the L–R sum-maximiser is $(0.6, 0.6)$ (midpoint of L and R on each dimension). Then $U_L = U_R = -0.5[(0.4)^2 + (0.4)^2] = -0.16$, vs. -0.17 at the default.

So logrolling *still* produces a (small) gain — *salience asymmetry is not necessary.* What it is, is an *amplifier*: the asymmetric case yields gains of $0.202 - 0.1344 = 0.068$ per party, while the symmetric case yields only $0.17 - 0.16 = 0.01$. The salience case opens up large gains-from-trade because each party can concede heavily on its low-salience dimension at low cost while extracting big gains on its high-salience one.

2.3 Mapping the Pareto-improving region (simulation)

Listing 2: Pareto-region map

```
grid <- expand.grid(x = seq(0, 1, length.out = 200),
                    y = seq(0, 1, length.out = 200))
aL <- 0.3; aR <- 0.7
grid$UL <- -(aL*(grid$x - 0.2)^2 + (1 - aL)*(grid$y - 1.0)^2)
grid$UR <- -(aR*(grid$x - 1.0)^2 + (1 - aR)*(grid$y - 0.2)^2)
UL_def <- -0.202; UR_def <- -0.202
grid$pareto <- with(grid, UL > UL_def & UR > UR_def)

library(ggplot2)
ggplot(grid, aes(x, y, fill = pareto)) + geom_raster() +
  geom_point(aes(x = 0.5, y = 0.5), inherit.aes = FALSE, colour = "red") +
  geom_point(aes(x = 0.76, y = 0.76), inherit.aes = FALSE, colour = "green") +
  coord_equal() + theme_minimal()
```

(a)–(c): expected output. The Pareto region is a lens-shaped set in the centre of the unit square, symmetric about the diagonal. The default $(0.5, 0.5)$ sits on its boundary; the logroll-optimal $(0.76, 0.76)$ sits well inside. Sweeping α_L over $\{0.1, 0.3, 0.5, 0.7, 0.9\}$ (with $\alpha_R = 1 - \alpha_L$):

α_L	Region area (fraction)	Diagonal logroll-optimal x^*
0.1	0.223 (largest)	0.92
0.3	0.123	0.76
0.5	0.006	0.60
0.7	0.002 (smallest)	0.44
0.9	0.162	0.28

The pattern is *non-monotone* in α_L: extreme salience asymmetries ($\alpha_L \in \{0.1, 0.9\}$) produce the largest Pareto regions, while the case $\alpha_L = 0.7$ produces a region *smaller* than the no-asymmetry case $\alpha_L = 0.5$. The reason: at $\alpha_L = 0.7$ the asymmetry has L caring most about the issue where it is *closest* to the default (x, distance 0.3), and analogously for R. The small-distance dimension dominates each party's utility, so neither has much pain at the default and there is little to gain. The largest region ($\alpha_L = 0.1$) instead has each party caring most about the dimension where it is *farthest* from the default — both pain and gain-potential are maximised.

(d) Spatial setup with three legislators. With $\alpha = 1/2$, ideal points $L = (0.2, 1.0), M = (0.5, 0.5), R = (1.0, 0.3)$. The L–R Pareto-improving region (vs. $U_L^{\text{MR}} = -0.34$, $U_R^{\text{MR}} = -0.29$) is non-empty and contains the slide bundle. The sub-region in which M is also better off than at $(0.5, 0.5)$ is empty: any improvement for both L and R relative to the median outcome moves the bundle off the centre, harming M. So a true Pareto improvement does not exist. This is why M blocks the deal in the absence of party discipline.

(e) Take-away. The simulation makes vivid that "gains from trade" between L and R are mediated by their salience asymmetry, and that the harmed party (M) is precisely the median legislator whose vote would otherwise determine the outcome. The institutional function of disciplined parties is thus not just to capture the L–R coalition's gains but to *commit* the coalition to ignoring M's veto.

2.4 Logrolling, gridlock, and credibility (essay sketch)

Argument structure.

(a) The unravelling argument: in a one-shot legislature without parties, legislators face a hold-up problem. After L has voted in favour of R's preferred policy, R has no incentive to honour the deal on L's preferred policy. Anticipating this, L refuses to vote first. By backward induction the coalition unravels. Parties solve the problem by punishing defectors via list position, committee assignment, etc.

(b) Two-period gridlock. Imagine two issues, A and B, in two periods. Period 1 faces a Pareto-improving but *politically symbolic* reform on issue A (everyone gains). Period 2 will face issue B, on which L and R disagree sharply. If R blocks the period-1 reform, R may extract concessions on issue B in exchange for finally allowing it. So strategic delay can dominate immediate adoption even when adoption is unanimous-utility-improving. A short payoff matrix makes the point: option 1 (adopt A now, fight B) vs. option 2 (block A, bundle later) — the second can dominate for R.

(c) Are Levy and Lee compatible? Yes. Levy says parties *enable* coalitional commitment; Lee says they enable *strategic* bundling, which can be welfare-reducing. Empirically: Levy's logic dominates when (i) issues are many and intensities asymmetric (gains from trade); (ii) the time horizon is short or issues uncorrelated (bundling has limited strategic value). Lee's logic dominates when (i) issues are few and salience high; (ii) horizon is long (delay valuable); (iii) party discipline is strong enough to enforce a hold-up.

(d) Application: e.g. US Congress, where divided government and committee gatekeeping can produce strategic delay (Lee), but where party discipline on logroll-style omnibus packages (e.g. infrastructure plus reconciliation) also produces Levy-style gains.

3 Sunk costs, signalling, and the loyalty–competence trade-off

3.1 The separating equilibrium

Throughout this section, $\bar{\alpha}$ denotes the high competence type and $\underline{\alpha}$ the low.

(a) Agent-side ICs. High type, given that $a(1) = 1$ and $R \geq w(\bar{\alpha})$:

$$\text{IC-H:} \qquad \underbrace{R - \kappa(\bar{\alpha})}_{\text{exert effort, get promoted, accept}} \quad \geq \quad \underbrace{0}_{\text{don't, not promoted, fall back to nothing}} \quad \iff R \geq \kappa(\bar{\alpha}).$$

Low type, deviation to $e = 1$ would yield $R - \kappa(\underline{\alpha})$ if promoted (since $w(\underline{\alpha}) < w(\bar{\alpha}) \leq R$):

$$\text{IC-L (no-mimic):} \quad 0 \;\geq\; R - \kappa(\underline{\alpha}) \iff R \leq \kappa(\underline{\alpha}).$$

Combining: $\boxed{\kappa(\bar{\alpha}) \leq R \leq \kappa(\underline{\alpha})}$.

(b) Party-side conditions. After observing $e = 1$, posterior $\mu(\bar{\alpha} \mid 1) = 1$. Party promotes iff $v\bar{\alpha} \geq K$:

$$\text{Promote on } e = 1: \quad K \leq v\bar{\alpha}.$$

After observing $e = 0$, posterior $\mu(\bar{\alpha} \mid 0) = 0$. Party promotes iff $v\underline{\alpha} \geq K$, i.e. does *not* promote iff $K > v\underline{\alpha}$:

$$\text{Don't promote on } e = 0: \quad K > v\underline{\alpha}.$$

Combining: $\boxed{v\underline{\alpha} < K \leq v\bar{\alpha}}$.

(c) Off-path beliefs and PBE check. Off-path belief: $\mu(\bar{\alpha} \mid e = 0) = 0$ (or any belief $\leq K/(v\bar{\alpha})$ that prevents promotion). With this belief, the proposed $a(0) = 0$ is a best response. The high type plays $e = 1$ optimally (IC-H); the low type plays $e = 0$ optimally (IC-L). On-path beliefs follow Bayes' rule. So this constitutes a PBE.

(d) Comparative statics in R. R enters two constraints with opposite signs:

$$\text{IC-H requires } R \geq \kappa(\bar{\alpha}); \quad \text{IC-L requires } R \leq \kappa(\underline{\alpha}).$$

Increasing R relaxes IC-H but tightens IC-L. The separating region in R is the closed interval $[\kappa(\bar{\alpha}), \kappa(\underline{\alpha})]$. Above the upper bound, the low type mimics; below the lower bound, the high type shirks. Either failure produces a pooling equilibrium.

(e) Outside option of the high type. The maintained assumption $R \geq w(\bar{\alpha})$ becomes the binding lower bound on R when $w(\bar{\alpha}) > \kappa(\bar{\alpha})$. As $w(\bar{\alpha})$ rises, the effective lower bound moves up, shrinking the separation interval to $[\max\{\kappa(\bar{\alpha}), w(\bar{\alpha})\}, \kappa(\underline{\alpha})]$. When $w(\bar{\alpha}) > \kappa(\underline{\alpha})$, separation breaks entirely: even the maximum reward consistent with no-mimicry fails to secure retention. This is the formal counterpart to the empirical claim that parties struggle to attract talent in tight private-sector labour markets.

3.2 Pooling and partial pooling

(a) Pooling on $e = 1$. Both types choose $e = 1$. On-path posterior $\mu(\bar{\alpha} \mid 1) = p_H$ (the prior). Party promotes after $e = 1$ iff $v\,\mathbb{E}[\alpha] \geq K$, i.e.

$$v(p_H\bar{\alpha} + (1 - p_H)\underline{\alpha}) \;\geq\; K.$$

Off-path posterior: $\mu(\bar{\alpha} \mid 0)$ chosen to make $a(0) = 0$ a best response (for instance $\mu(\bar{\alpha} \mid 0) = 0$).

For both types to choose $e = 1$ in equilibrium, deviation to $e = 0$ must yield weakly less:

$$\text{IC for type } \alpha: \quad -\kappa(\alpha) + R \geq 0 \iff R \geq \kappa(\alpha).$$

The binding constraint is the low type: $R \geq \kappa(\underline{\alpha})$. Thus pooling on $e = 1$ is sustainable iff $R \geq \kappa(\underline{\alpha})$ *and* $v\,\mathbb{E}[\alpha] \geq K$.

(b) Pooling on $e = 0$. Both types choose $e = 0$. On-path posterior is the prior p_H. The party promotes iff $v\,\mathbb{E}[\alpha] \geq K$. If $v\,\mathbb{E}[\alpha] < K$, the party does not promote after $e = 0$.

For both types to choose $e = 0$ rather than deviate, the deviation must yield weakly less. With off-path belief $\mu(\bar{\alpha} \mid 1)$ that supports $a(1) = 0$ (e.g. $\mu = 0$), deviation gives $-\kappa(\alpha) < 0$ — clearly worse. So pooling on $e = 0$ is sustainable iff

$$\boxed{v\,\mathbb{E}[\alpha] = v(p_H\bar{\alpha} + (1 - p_H)\underline{\alpha}) < K.}$$

Plain language: when scarce-position cost is high enough that the party wouldn't promote even given the prior, no agent has any incentive to incur κ. Effort collapses; the screening device is dead.

(c) Partial pooling. Strategy: $s(\bar{\alpha}) = 1$, $s(\underline{\alpha}) = 1$ with probability $q \in (0, 1)$, otherwise 0. Bayes' rule:

$$\mu(\bar{\alpha} \mid 1) = \frac{p_H}{p_H + (1 - p_H)q}, \qquad \mu(\bar{\alpha} \mid 0) = 0$$

(off-path or unused, since high type never chooses $e = 0$ on path).

For low type's mixing to be incentive-compatible, she must be indifferent. Let $\rho \in (0, 1)$ be the party's promotion probability after $e = 1$:

$$\text{Indifference for low type:} \quad \rho R - \kappa(\underline{\alpha}) = 0 \iff \rho^* = \kappa(\underline{\alpha})/R.$$

For the party to mix, it must be indifferent between promoting and not after $e = 1$:

$$\mu(\bar{\alpha} \mid 1)v\bar{\alpha} + (1 - \mu(\bar{\alpha} \mid 1))v\underline{\alpha} \;=\; K \iff \mu^* \;=\; \frac{K/v - \underline{\alpha}}{\bar{\alpha} - \underline{\alpha}}.$$

Combining with the Bayes' rule expression and solving for q:

$$q^* \;=\; \frac{p_H(1 - \mu^*)}{(1 - p_H)\mu^*}.$$

(d) Bénabou–Tirole behavioural channel. Even if the type composition of agents is fixed, raising R in a richer model (where actions in office signal type) erodes the informational content of those actions. When pay is modest, only intrinsically motivated types find the office attractive; observable prosocial behaviour by an officeholder is therefore a strong signal of motivation. With high R, opportunists pool with the intrinsically motivated, and the same observable behaviour is no longer diagnostic. The signal-dilution effect arises from *behaviour*, not *selection*: it operates through the link between R and the agent's incentive to produce informative on-the-job actions, not through the entry margin.

3.3 Mapping the equilibrium space (simulation)

Listing 3: Equilibrium classification on (R, κ_{low}) grid

```
classify <- function(R, kL, kH = 1, K = 1.4, v = 1, aL = 1, aH = 2,
                     wH = 0.8, pH = 0.5) {
  sep <- (kH <= R) && (R <= kL) && (v*aL < K) && (K <= v*aH) && (R >= wH)
  pool1 <- (R >= kL) && (v*(pH*aH + (1-pH)*aL) >= K)
  pool0 <- (v*(pH*aH + (1-pH)*aL) < K)
  if (sep && pool1) "both"
  else if (sep) "separating"
  else if (pool1) "pool e=1"
  else if (pool0) "pool e=0"
  else "neither"
}
```

(a)–(b) Partition. With $\kappa_H = 1$, $K = 1.4$, $v = 1$, $\underline{\alpha} = 1$, $\bar{\alpha} = 2$, $p_H = 0.5$: party-side condition $v\underline{\alpha} = 1 < K = 1.4 \leq v\bar{\alpha} = 2$ holds, and $v\,\mathbb{E}[\alpha] = 1.5 > K = 1.4$ rules out pool-on-$e = 0$. The partition is determined by the agent-side bounds:

- $R < \kappa_H = 1$: high type's IC fails. No separating; if $R \geq \kappa_L$ also fails, no pure equilibrium with effort. (Pool on $e = 0$ remains unavailable at this K.)
- $1 \leq R \leq \kappa_L$: separating exists. The party-side condition holds and the maintained $R \geq w(\bar{\alpha}) = 0.8$ holds.
- $R > \kappa_L$: low type mimics. Pool on $e = 1$ takes over (sustainable since $v\,\mathbb{E}[\alpha] \geq K$).

Plot: a vertical strip of "separating" between $R \in [1, \kappa_L]$, widening linearly with κ_L, and a "pool $e = 1$" wedge above the strip.

(c) Vary K.

- $K = 0.8 < v\underline{\alpha} = 1$: party-side condition fails (party would promote even after $e = 0$). Separating breaks down; the relevant equilibrium has both types choosing $e = 0$ (no need for effort, since promotion is automatic).
- $K = 1.4$: as in (a). Separating + pool $e = 1$ partition the plane.
- $K = 1.6 > v\,\mathbb{E}[\alpha] = 1.5$ but $\leq v\bar{\alpha} = 2$: pool on $e = 0$ becomes sustainable everywhere (since the party would not promote even on the prior). Separating still exists in the agent-IC strip and now *coexists* with pool on $e = 0$.
- $K = 2.5 > v\bar{\alpha}$: even the high posterior cannot justify promotion. Pool on $e = 0$ is the unique equilibrium everywhere.

The thresholds: $K = v\underline{\alpha} = 1$ (below: party-side condition fails); $K = v\,\mathbb{E}[\alpha] = 1.5$ (above: pool-$e = 0$ becomes available); $K = v\bar{\alpha} = 2$ (above: separating breaks because the party won't promote even given high posterior).

(d) Vary $w(\bar{\alpha})$. As $w(\bar{\alpha}) \in [0, 3]$ rises with $R = 1.5$ fixed, the separating region holds until $w(\bar{\alpha}) = 1.5$, after which the maintained assumption $R \geq w(\bar{\alpha})$ fails: separation collapses because the high type would reject the promotion. This recovers §3.1(e).

(e) Partial-pooling simulation. For one parameter setting in the partial-pooling region, simulate $N = 1000$ agents drawn from the prior $p_H = 0.5$. Each high type plays $e = 1$. Each low type plays $e = 1$ with probability q^*. The party promotes after $e = 1$ with probability ρ^*. Then the share of promoted agents who are competent is

$$\Pr(\bar{\alpha} \mid \text{promoted}) = \mu^*.$$

Compare with separating (1.0) and $e = 0$ pooling (no one is promoted, so the share is undefined). Welfare-cost interpretation: partial pooling produces a posterior between the prior and 1, but at the cost of (i) lost effort by some low types and (ii) wasted promotions on those low types who randomised in.

3.4 From model to institutions (essay sketch)

(a) Two tools. Sample answer: *seniority-based list rules* primarily affect the *reward R over time*, by making promotion conditional on having served for many years (a delayed, accumulated reward). *Forced severing of outside ties* (e.g. MPs banned from second jobs, residency requirements) primarily affects the *outside option* $w(\alpha)$, mechanically shrinking w for all types and disproportionately for high-a types whose outside options are larger in absolute terms. The same tool can plausibly affect more than one primitive: the seniority rule also raises the effective effort cost $\kappa(\alpha)$ if early-career service requires unpaid work.

(b) Cirone, Cox, and Fiva (2020) on seniority. The empirical pattern — long-standing loyalty rewarded with safer list positions — is most consistent with a *separating* equilibrium in which κ is delivered as a sequence of effort instalments and R is a delayed prize. The mechanism does not need to invoke pooling. (However: parties also use seniority in ways consistent with Mattozzi–Merlo career models, where retention is the prize.)

(c) Buisseret–Prato two-tier rule. Sketch: the party offers two list-slot types. *Safe slots* pay a high R_{safe} but require a high effort prerequisite κ_{safe}, attracting loyal types whose $w(\alpha)$ is low. *Marginal slots* pay $R_{\text{marg}} < R_{\text{safe}}$ but are conditional on observable performance in office, attracting talent willing to gamble on outcomes. Formally: the agent chooses which slot to apply for, the party screens by self-selection. The two-tier rule *separates by motivation* (loyal vs. talented), not by competence per se.

(d) Advice to the centrist party. *Plausible changes:* (i) raise R for centrist candidates (e.g. via guarantees of cabinet posts or list rank); (ii) lower κ on early-career expectations (less unpaid party work); (iii) institutionalise pathways from private-sector careers to politics (lateral entry).

Backfires: raising R aggressively triggers low-type mimicry; lowering κ undermines screening; lateral entry programmes that bypass party-track costs may dilute the brand. The fundamental trade-off — between screening for loyalty and retaining talent — is not solved by "trying harder" on any one margin; it requires the party to either (i) accept lower retention of high-w types, or (ii) tolerate weaker screening on m.

4 Why parties exist: brands, teams, transaction costs, and the politicians' wage

4.1 Moral hazard in electoral teams

(a) Equal sharing. Each candidate maximises

$$U_i(e_i; e_{-i}) = \tfrac{1}{N}\,\alpha(e_i + \textstyle\sum_{j\neq i} e_j) - \tfrac{1}{2}e_i^2.$$

FOC: $\alpha/N - e_i = 0 \Rightarrow e_i^* = \alpha/N$. By symmetry, total team effort is $\sum e_j^* = \alpha$.

Planner's problem: maximise $\alpha \sum e_j - \frac{1}{2}\sum e_j^2$. FOC: $\alpha - e_i = 0 \Rightarrow e_i^{\text{FB}} = \alpha$. Total: $N\alpha$.

Comparison: $e^*/e^{\text{FB}} = 1/N$. The wedge widens as $N \to \infty$ because each candidate internalises only $1/N$ of marginal team output. Free-riding scales with team size.

(b) Rank-based with $\phi(r) = \beta/r$.

$$U_i = (\beta/r_i)\,\alpha\big(\textstyle\sum_j e_j\big) - \tfrac{1}{2}e_i^2.$$

FOC: $\alpha\beta/r_i = e_i \Rightarrow e_i^* = \alpha\beta/r_i$, with $\partial e_i^*/\partial r_i = -\alpha\beta/r_i^2 < 0$.

(c) Numerical comparison. Equal sharing: total effort $= N \cdot (\alpha/N) = \alpha$, *independent of* N.

Rank-based: total effort $= \alpha\beta \sum_{r=1}^{N} 1/r = \alpha\beta H_N$, where H_N is the N-th harmonic number. With $\alpha = \beta = 1$:

N	Equal-sharing total	Rank-based total (H_N)
2	1.000	1.500
5	1.000	2.283
10	1.000	2.929
50	1.000	4.499

The rank-based scheme dominates equal sharing for all $N \geq 2$. The advantage grows with N because rank-based rewards convert what would be free-riders into top-of-list strivers.

(d) Closed vs. open list. *Closed list:* party fixes $\phi(r)$ pre-election; each candidate's reward function is common knowledge and stable. Top of list has the highest ϕ and the highest e. *Open list:* voters reshuffle ranks based on their inferences about candidate effort. ϕ becomes endogenous and depends on voter beliefs.

Under what conditions does open list elicit *less* effort? When voter signals about effort are noisy and voters cannot discriminate well, the expected ϕ each candidate faces becomes uniform across candidates (all expect the same average rank). This blunts the rank-based incentive and pushes total effort toward the equal-sharing benchmark. So closed list dominates open list when the signal environment is poor.

4.2 Party brands as informational shortcuts

(a) Posterior derivation. Standard normal-normal Bayesian update. Prior: $\theta_j \sim \mathcal{N}(\mu_P, \sigma^2)$. Likelihood: $z_j \mid \theta_j \sim \mathcal{N}(\theta_j, \tau^2)$. Posterior:

$$\theta_j \mid z_j, P \sim \mathcal{N}\left(\frac{\tau^{-2}z_j + \sigma^{-2}\mu_P}{\tau^{-2} + \sigma^{-2}}, \frac{1}{\tau^{-2} + \sigma^{-2}}\right).$$

Hence $\mathbb{E}[\theta_j \mid z_j, P]$ is the precision-weighted average.

(b) Comparative statics. Weight on μ_P is $\sigma^{-2}/(\tau^{-2} + \sigma^{-2})$.

- $\tau^2 \to \infty$ (uninformative signals): $\tau^{-2} \to 0$, so the brand weight goes to 1. Plain language: when you cannot learn from z_j, you fall back on the prior μ_P.
- $\sigma^2 \to 0$ (homogeneous parties): $\sigma^{-2} \to \infty$, so the brand weight goes to 1. Plain language: when the candidate barely deviates from the brand, the brand *is* the candidate's policy.

(c) Policy estimation MSE. The MSE of the signal-only estimator is $\mathbb{E}[(z_j - \theta_j)^2] = \tau^2$ (measurement noise variance). The MSE of the Bayesian shrinkage estimator equals the posterior variance $1/(\tau^{-2} + \sigma^{-2}) = \tau^2\sigma^2/(\tau^2 + \sigma^2)$. The MSE *reduction* from using the brand is

$$\Delta\text{MSE} = \tau^2 - \frac{\tau^2\sigma^2}{\tau^2 + \sigma^2} = \frac{\tau^4}{\tau^2 + \sigma^2}.$$

This is strictly positive for $\tau, \sigma > 0$, increasing in τ, and *decreasing* in σ. Brand-aided shrinkage helps most when signals are noisy and parties are homogeneous.

Listing 4: Policy-estimation MSE comparison (R)

```
sim_brand_mse <- function(tau, sigma = 0.05, nA = 1000, nB = 1000,
                          muA = 0.3, muB = 0.7) {
  thA <- rnorm(nA, muA, sigma); zA <- thA + rnorm(nA, 0, tau)
  thB <- rnorm(nB, muB, sigma); zB <- thB + rnorm(nB, 0, tau)
  z <- c(zA, zB); theta <- c(thA, thB); mu <- c(rep(muA, nA), rep(muB, nB))
  hat_S <- z
  hat_B <- (z / tau^2 + mu / sigma^2) / (1 / tau^2 + 1 / sigma^2)
  c(MSE_signal = mean((hat_S - theta)^2),
    MSE_bayes  = mean((hat_B - theta)^2),
    theory_red = tau^4 / (tau^2 + sigma^2))
}
sapply(c(0.025, 0.05, 0.1, 0.2), sim_brand_mse)
```

Note on classification. A common impulse is to test the brand by asking whether voters can correctly *classify* candidates by party. Under symmetric priors and equal variances the optimal Bayesian classifier collapses to the midpoint rule $z \lesseqgtr (\mu_A + \mu_B)/2$, so the brand provides no classification value. The brand's informational role is in *policy estimation*, not classification — this is the substance of Snyder–Ting's argument.

(d) Lupu (2016). "Brand dilution" in the model maps to one of: μ_P drift over time (parties abandon historical positions), σ^2 rising (the party becomes a coalition of heterogeneous factions), or both. Predicted empirical implications:

- Higher electoral volatility (voters lose their informational anchor).
- Increased reliance on personal candidate signals (the z_j).
- Asymmetric vulnerability: parties whose original brand was sharp suffer more from drift than parties whose brand was always vague.

4.3 The Coasean rationale (essay sketch)

Argument structure.

(a) *Coasean translation.* The "transaction" is a sustained agreement to support each other's policies across time and across issues. Without parties, each new majority would be assembled de novo, with negotiation, monitoring, and enforcement costs. Parties internalise these costs: legislators are pre-committed to vote together (whips, expulsion), so the marginal cost of assembling a majority on any issue is approximately zero. This is the legislative analogue of Coase's argument for the firm: when transaction costs of repeated contracting are high, integration into a hierarchy dominates.

(b) *Presidential vs. parliamentary.* Coasean view predicts that parliamentary systems — where legislative-executive fusion makes party discipline self-reinforcing — should have stronger, more cohesive parties; presidential systems with separation of powers reduce the party's monopoly on legislative coordination, weakening discipline. Levy's commitment-problem view predicts similarly: parliamentary systems generate more multidimensional bargaining demands, which require party-level commitment. The two views are largely co-extensive on this margin, but Levy specifically requires *multiple issue dimensions* for parties to add value, while Coase only requires *repeated transactions*. Single-issue legislatures can have parties under Coase but not under Levy.

(c) *Coase vs. Lee compatibility.* Both can be true. Coase explains why parties exist; Lee explains why *the same parties* can also produce gridlock by strategically withholding transactions. Conditions favouring Coase: many issues, low intensity asymmetry, short horizon, weak-to-moderate party discipline. Conditions favouring Lee: few issues, high intensity asymmetry, long horizon, strong party discipline.

(d) *Application.* A reasonable answer chooses, e.g., the EU Parliament (Coasean: party groups reduce coordination costs across 27 national delegations; Lee gridlock: bundling of unrelated dossiers in trilogue) and assesses where each story bites.

4.4 Should politicians be paid more? (essay sketch)

(a) Decomposition. Differentiating $\mathbb{E}[q \mid S(W)] = \bar{a}(W) + (1-\rho)\beta\bar{m}(W)$ with respect to W:

$$\frac{d\mathbb{E}[q \mid S(W)]}{dW} = \frac{d\bar{a}(W)}{dW} + (1-\rho)\beta\,\frac{d\bar{m}(W)}{dW}.$$

The first term (*competence channel*) is positive: at the entry margin, raising W admits citizens with higher $w(a)$, who tend to have higher a since $w'(a) > 0$. The second term (*morality channel*) is sign-ambiguous: the marginal entrant induced by a higher W has a lower m (because they were not willing to enter at the lower W, which is consistent with low intrinsic motivation), but the magnitude depends on the joint distribution.

(b) Joint-distribution conditions. The sign of $d\bar{m}/dW$ depends on the conditional density of m on the entry frontier. Two clean limits:

- $\mathrm{corr}(a, m) > 0$ in the population: marginal entrants are high-a *and* high-m. Both channels are positive — the *double-dividend* case.
- $\mathrm{corr}(a, m) < 0$ (or zero): marginal entrants are high-a low-m rent-seekers. The second channel is negative; the net sign depends on $(1-\rho)\beta$ vs. the slope of $\bar{a}(W)$.

(c) Connection to party competition. Consider the Downsian extension. Voters care about platform proximity and quality q_j. Party j's vote share becomes

$$V_j = \Pr\big(-|x_i - x_j| + \gamma q_j(W_j) > -|x_i - x_{-j}| + \gamma q_{-j}(W_{-j})\big),$$

for some weight $\gamma > 0$ on quality.

(i) Each party's marginal incentive in setting W_j is to raise q_j in order to attract voters, balanced against the budget constraint or against any internal cost of high W (e.g. public backlash).

(ii) Downsian convergence on x *does* extend to convergence on W in symmetric settings: both parties race to the top on quality if quality is unambiguously good. But *convergence on W fails* if (a) the parties' candidate pools differ (so the same W produces different q), or (b) the morality channel is sign-ambiguous and a party with strong brand prefers to keep W low to preserve self-selection by motivation.

(iii) One party may differentiate on W if its core constituency rewards ideological purity (low W retains motivated true believers) while the other's rewards generic competence (high W attracts professional politicians). This predicts an empirical correlation between party brand strength and average political wage.

(d) Empirical literature. Ferraz & Finan (2009), Gagliarducci & Nannicini (2013): higher pay $\Rightarrow$ more educated and more experienced legislators ($\mathrm{d}\bar{a}/\mathrm{d}W > 0$). Kotakorpi & Poutvaara (2011): heterogeneous selection effect — large for women, small for men. Carnes & Hansen (2016): higher pay does *not* broaden the class background of legislators.

What this says about the model:

- The empirical pattern $\mathrm{d}\bar{a}/\mathrm{d}W > 0$ is consistent with either (i) $\mathrm{corr}(a, m) > 0$ in those settings, (ii) high ρ (strong accountability eliminates the morality channel), or (iii) β small in the welfare function. The data do not separate these three; this is a textbook identification problem.
- The morality channel is essentially unmeasured: existing studies use education, occupation, and performance as proxies for a, not for m. Direct measurement of m would require survey or behavioural data on motivations and ethical conduct.
- The optimal political wage is therefore not pinned down by the data we have. It is a *useful concept* but a *poor policy target*: most margins (recruitment, party brand, electoral institutions) probably dominate the wage margin in explaining cross-country differences in legislator quality.

Final take. A confident answer notes: the question "should politicians be paid more?" becomes well-defined only after stipulating ρ and $\mathrm{corr}(a, m)$. In high-ρ, high-corr settings (some Scandinavian democracies?), raise the wage. In low-ρ, low-corr settings (clientelistic systems with weak audits), pay raises will likely select for opportunists rather than reformers. The empirical literature is more optimistic than the theory because it identifies the easy-to-measure half of the question.